ASSESSMENT REMIT

Name			
ID#			
Semester	Fall 2026	**Section**	
School	School of Engineering, Architecture and Interior Design (EAID)		
Program	Bachelor of Science in Information Technology		
Major	Information Technology		
Course Code	**ITEC 310**		
Course Title	Internet of Things		
Level	☐ Remedial ☐ Bridging ☑ Undergraduate ☐ Graduate		
Assessment Title	Group Lab Project — AI-Assisted IoT Application Development		
Weighting	15 Marks		
Instructor	Dr. Maher Alrahhal		
Hand Out Date	02/11/2026		
Due Date & Time	Date:03/12/2026	Time: 1:00 – 2:50 PM	
Feedback Post Date	14/12/2026		
Assessment Mode	☐ Individual ☑ Group ☐Closed Book ☑ Open Book		
Assignment Length *(for assignment-based assessments)*	Maximum five-page report + working application		
Submission Mode	☐ Online by Email ☑ Online through LMS | TurnItIn ☐ Hard Copy		
Document Type	☐ Words ☑ PDF ☐ Artwork ☑ Software ☐ Other		

GENERAL REQUIREMENTS (GROUP PROJECT)

- Each group must have 3 or 4 students (maximum 4 students per group). Groups with fewer than 3 or more than 4 students will not be accepted.
- Each group must select one IoT-related problem and develop a simple working website or mobile application.
- Students must use Visual Studio Code (VS Code) with an AI coding assistant such as OpenAI Codex or Claude, or use Google Antigravity.
- The application must demonstrate at least one IoT concept involving sensors, monitoring, communication, or automatic control.
- Physical IoT hardware is not required. Students may use simulated sensor data.
- Each group must submit a GitHub repository containing the complete source code and a README file.
- Each group must submit a short PDF report (maximum five pages).
- Each group must demonstrate the working application and participate in a technical discussion.
- All group members must understand the application and explain their individual contributions.
- AI tools may assist with coding, but students are responsible for testing, understanding, and explaining their work.
- Academic integrity and university assessment policies must be followed.

ASSESSMENT:

Project Selection

Each group must select ONE (1) IoT project from the following options:

1. Smart Home Monitoring: Monitor room temperature and control a virtual light or fan.
2. Smart Parking System: Display parking spaces as available or occupied and calculate available spaces.
3. Smart Agriculture: Monitor simulated soil moisture and automatically control a virtual water pump.
4. Smart Classroom: Monitor classroom temperature and automatically activate a virtual fan when the temperature exceeds 30°C.
5. Smart Security System: Detect simulated motion and display security alerts.
6. Smart Energy Monitoring: Monitor electricity consumption for three appliances and calculate total consumption.

Students may propose another IoT-related project with instructor approval.

Requirements: Each group must develop a working website or mobile application using AI-assisted coding. Physical IoT hardware is optional, and simulated sensor data is acceptable.

All groups must complete the four assessment tasks (Tasks 1–4), regardless of the selected project.

Complete ALL FOUR (4) assessment tasks.

Task 1 – Working IoT Application (6 Marks)

Develop a simple working IoT website or mobile application using AI-assisted coding.

The application must:

a) Run correctly and provide a working user interface.

b) **Select the right sensors** for the **select**ed IoT application and display simulated or real sensor data, such as temperature, soil moisture, distance, motion, or energy consumption.

c) **Apply IoT concepts** to **solve problems** by performing an appropriate IoT action, such as activating a virtual fan, controlling a water pump, displaying an alert, or updating a parking status.

d) **Demonstrate** the **basic configuration** related to **IoT sensors**, including how sensor readings are generated or updated, to accomplish **group tasks**.

Suggested Projects:

- Smart Home Monitoring
- Smart Parking System
- Smart Agriculture
- Smart Classroom
- Smart Security System

- Smart Energy Monitoring

Students may propose another suitable IoT project with instructor approval.

Task 2 – Project Documentation (4 Marks)

Prepare a PDF report of no more than five pages.

The report must include:

a) Project Overview: Project title, objective, and a brief explanation of how the application works.

b) AI Development: AI coding assistant used and 3–5 actual prompts, with brief explanations.

c) **Write the lab report** and **present the output result** as **lab record**, including screenshots of sensor readings and system responses.

d) Conclusion: Briefly explain whether the application achieved its objective.

Task 3 – GitHub Repository (2 Marks)

Create a GitHub repository containing:

a) The complete working source code.

b) A README file explaining the project and how to run it.

c) The project report.

Submit the GitHub repository link.

Task 4 – Individual Project Demonstration and Discussion (3 Marks)

Each group must demonstrate the application and participate in a discussion.

Students must:

a) **Demonstrate** the working application and the **basic configuration** related to **IoT sensors** to accomplish **group tasks**.

b) **Select** and explain **the right sensors** and **communication protocols** used to accomplish the **group based task** in the IoT environment.

c) **Apply IoT concepts** and **communicate the technical aspects** of the application, including how AI coding tools were used to **solve problems**.

All group members must participate individually in the discussion (3 marks).

ANSWER GUIDELINES:

Students are expected to:

- Complete all FOUR (4) assessment tasks.
- Develop one working IoT website or mobile application.
- Demonstrate the use of simulated or real sensor data.
- **Select the right sensors** and **communication protocols** and explain the system operation and automatic control logic.

- **Apply IoT concepts** and **communicate the technical aspects** to **solve problems**; test the application and explain the results.
- Publish the complete source code on GitHub.
- Prepare a PDF report of no more than five pages.
- **Demonstrate** the **basic configuration** related to **IoT sensors** and participate individually in the discussion (3 marks).

Presentation Requirements

- Include all group members' names and student IDs.
- Clearly identify the selected IoT project.
- Include screenshots of the working application.
- Include 3–5 actual AI prompts used during development.
- Provide the GitHub repository link.
- Explain each group member's contribution.
- Use clear technical explanations and appropriate references.

REFERENCING STYLE:

Students should use the American Psychological Association style (APA) as per the following guidelines: https://apastyle.apa.org/style-grammar-guidelines/references/examples

PLAGIARISM & GENERATIVE AI (FOR ASSIGNMENT BASED ASSESSMENTS):

Amity University Dubai is committed to academic honesty and integrity.
For this project, students are permitted and required to use at least one AI coding assistant, such as OpenAI Codex, Claude, or Google Antigravity, to support application development.
Students must:

- Document the AI tools and actual prompts used.
- Review and test all AI-generated code.
- Understand and explain the application and its IoT functionality.
- Write their own project analysis, results, and conclusions.
- Acknowledge external sources and AI assistance appropriately.

Submitting work that students cannot explain or presenting AI-generated material as entirely independent work may constitute academic misconduct under university policy.

LATE SUBMISSIONS (FOR ASSIGNMENT BASED ASSESSMENTS):

You are advised to strictly adhere to the deadline. While Amity University Dubai might consider extenuating circumstances to support students who are experiencing unforeseen issues impacting their ability to complete assessments on time, any anticipated submission delay must be communicated in advance to the instructor who will consult with the Program Leader and the Dean before a permission is granted. All permissions must be based on valid reasons backed up by concrete evidence. For more details, you are advised to consult the latest version of AMUD's Academic Integrity and Academic Misconduct Policy.

FEEDBACK PLAN:

The preparation for tasks included in this assessment will be supported by formative feedback during the class regular sessions and related office hours. Summative written feedback will be provided on the date indicated earlier.

ASSESSED COURSE LEARNING OUTCOMES:	**RELATED QUESTIONS**	**MAX SCORE**	**MARK**
CLO4. **Select the right sensors** and **communication protocols** to accomplish the **group based task** used in IoT environment.	Task 1	**3**	
CLO5. **Apply IoT concepts** and **communicate the technical aspects** in the labs to **solve problems**.	Task 1, Task 3	**5**	
CLO6. **Write the lab report** and **present the output result** as **lab record**.	Task 2	**4**	
CLO7. **Demonstrate** the **basic configuration** related to **IOT sensors** to accomplish **group tasks**.	Task 1, Task 4	**3**	
	TOTAL	**15**	

EMBEDDED PLO DIRECT ASSESSMENT QUESTIONS? ☑ **YES** ☐ **NO**
[IF YES, COMPLETE THE TABLE BELOW]

PLO	**EMBEDDED QUESTIONS**
PLO 1 – Analyze a complex computing problem and apply principles of computing and other relevant disciplines to identify solutions.	Task 1
PLO 2 – Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.	Task 1, Task 3
PLO 3 – Communicate effectively in a variety of professional contexts.	Task 2, Task 4
PLO 5 – Function effectively as a member or leader of a team, demonstrating leadership, collaboration, and project management in achieving objectives related to the discipline.	Task 4

	LEVEL OF ACHIEVEMENT

SCORING RUBRIC						
	DESCRIPTOR	**WEIGHT**	**4 [87%-100%]**	**3 [73%-86%]**	**2 [60%-72%]**	**1 [0%-59%]**
1	Working IoT Application (Task 1)	6	Fully functional application, correct sensor data, appropriate control action, and clear sensor configuration.	Mostly functional with minor issues.	Partially functional with incomplete IoT functionality.	Nonfunctional or missing application.
2	Project Documentation (Task 2)	4	Clear project overview, actual AI prompts, screenshots, results, and conclusion.	Mostly complete with minor omissions.	Incomplete explanations or missing evidence.	Missing or inadequate report.
3	GitHub Repository (Task 3)	2	Complete source code, README, and report uploaded correctly.	Mostly complete repository with minor omissions.	Incomplete source code or documentation.	Missing or unusable repository.
4	Demonstration and Discussion (Task 4)	3	Working demonstration; all members clearly explain the IoT functionality and AI usage.	Mostly clear demonstration and explanations.	Limited demonstration or incomplete explanations.	Unsuccessful demonstration or minimal understanding.

Student Declaration

By signing this cover page, you acknowledge the instructions above (general requirements) and accept all responsibilities resulting from breaching them as per the related University policies. I hereby declare that it is my original work and I have duly acknowledged and referenced the source of any quotes, thoughts or ideas incorporated in the work.

No.	Student Name	Student ID	Student Signature	Date
1				
2				
3				
4				

To be filled by faculty

FEEDBACK FROM FACULTY:

Faculty Name and Signature **Date**